Chapter 13

Section A: Multiple Choice

1. Business intelligence (BI) describes a comprehensive, cohesive, and integrated set of tools and processes used to:
   a) Capture, collect, integrate, store, and analyze data to support business decision making
   b) Process daily transactions only
   c) Replace operational databases
   d) Manage network hardware
2. The BI architecture is composed of many interconnected parts, including:
   a) People, processes, data, and technology
   b) Hardware, software, and networks
   c) Tables, rows, and columns
   d) Schedules, logs, and backups
3. Which component of the BI framework collects, filters, integrates, and aggregates internal and external data to be saved into a data store?
   a) Query and reporting
   b) ETL tools
   c) Data visualization
   d) Data monitoring and alerting
4. The data store in a BI framework is generally represented by a:
   a) Transaction log
   b) Data warehouse or data mart
   c) Network file server
   d) Spreadsheet
5. Which BI component performs data selection and retrieval and is used by the data analyst to create queries and reports?
   a) ETL tools
   b) Query and reporting
   c) Data visualization
   d) Data analytics
6. Which BI component presents data to the end user in a variety of meaningful and innovative ways, such as summary reports, maps, pie or bar graphs, and dashboards?
   a) Query and reporting
   b) Data visualization
   c) Data monitoring and alerting
   d) Data analytics
7. Which BI component allows real-time monitoring of business activities and can present concise information in a single integrated view?
   a) Data visualization

b) Query and reporting
c) Data monitoring and alerting
d) Data analytics

8. Which BI component performs data analysis and data-mining tasks using the data in the data store?
   a) Data analytics
   b) ETL tools
   c) Query and reporting
   d) Data visualization
9. The main goal of BI is:
   a) Improved decision making
   b) Faster transaction processing
   c) Reduced storage requirements
   d) Increased data redundancy
10. Which of the following is a benefit of BI?
    a) Integrating architecture
    b) Common user interface for data reporting and analysis
    c) Common data repository fostering a single version of company data
    d) All of the above
11. The precursor of the modern BI environment was the:
    a) Data warehouse
    b) First-generation decision support system (DSS)
    c) OLAP cube
    d) Data mart
12. A decision support system (DSS) is:
    a) A centralized database
    b) An arrangement of computerized tools used to assist managerial decision making
    c) A type of network hardware
    d) A transaction processing system
13. Operational data is mostly stored in:
    a) Multidimensional databases
    b) Relational databases
    c) Flat files
    d) NoSQL databases
14. Decision support data differs from operational data in which three main areas?
    a) Timespan, granularity, and dimensionality
    b) Size, speed, and cost
    c) Hardware, software, and networks
    d) Rows, columns, and tables

15. Which of the following is a characteristic of operational data?
    a) Historic data
    b) Summarized data
    c) Current operations and real-time data
    d) Non-normalized structures
16. Which of the following is a characteristic of decision support data?
    a) Atomic-detailed data
    b) Highly normalized
    c) Historic data and snapshots of company data
    d) High-update volumes
17. Operational data typically has:
    a) High granularity and broad time span
    b) Low granularity and narrow time span
    c) High summarization levels
    d) Complex multidimensional structures
18. Decision support data typically has:
    a) Low granularity and narrow time span
    b) High granularity and broad time span
    c) Mostly update transactions
    d) Simple to medium query complexity
19. Which of the following is a requirement of a decision support database?
    a) Database schema that supports complex data representations
    b) Data extraction and filtering tools
    c) Capability to support very large databases
    d) All of the above
20. A data warehouse is a(n):
    a) Integrated, subject-oriented, time-variant, nonvolatile collection of data that provides support for decision making
    b) Transactional database optimized for updates
    c) Temporary storage for operational data
    d) Type of network protocol
21. Which characteristic of a data warehouse means that it is a centralized, consolidated database that integrates data derived from the entire organization and from multiple sources with diverse formats?
    a) Subject-oriented
    b) Time-variant
    c) Integrated
    d) Nonvolatile
22. Which characteristic of a data warehouse means that data is arranged and optimized to provide answers to questions from diverse functional areas within a company?

a) Integrated
b) Subject-oriented
c) Time-variant
d) Nonvolatile

23. Which characteristic of a data warehouse means that data represents the flow of data through time?
    a) Integrated
    b) Subject-oriented
    c) Time-variant
    d) Nonvolatile
24. Which characteristic of a data warehouse means that once data enters the data warehouse, it is never removed?
    a) Integrated
    b) Subject-oriented
    c) Time-variant
    d) Nonvolatile
25. The process of extracting, transforming, and loading data into a data warehouse is known as:
    a) OLAP
    b) ETL
    c) DSS
    d) SQL
26. A small, single-subject data warehouse subset is called a:
    a) Data mart
    b) Fact table
    c) Dimension table
    d) Cube
27. A data mart is:
    a) A large, enterprise-wide data warehouse
    b) A small, single-subject data warehouse subset
    c) A type of operational database
    d) A network file server
28. A star schema is a data modeling technique used to:
    a) Map multidimensional decision support data into a relational database
    b) Normalize operational data
    c) Create transaction logs
    d) Manage network security
29. Which of the following is a component of a star schema?
    a) Facts
    b) Dimensions

c) Attributes
d) All of the above

30. In a star schema, numeric measurements that represent specific business aspects or activities are called:
    a) Dimensions
    b) Facts
    c) Attributes
    d) Hierarchies
31. In a star schema, qualifying characteristics that provide additional perspectives to a given fact are called:
    a) Facts
    b) Dimensions
    c) Metrics
    d) Keys
32. In a star schema, attributes are used to:
    a) Search, filter, or classify facts
    b) Store numeric measurements
    c) Define primary keys only
    d) Manage network connections
33. Attributes within dimensions can be ordered in a well-defined attribute hierarchy that provides:
    a) Top-down data organization
    b) Capability to perform drill-down and roll-up searches
    c) Both a and b
    d) None of the above
34. Online analytical processing (OLAP) is a BI style whose systems share which main characteristics?
    a) Multidimensional data analysis techniques
    b) Advanced database support
    c) Easy-to-use end-user interface
    d) All of the above
35. OLAP systems support:
    a) Decision making, business modeling, and operations research
    b) Daily transaction processing only
    c) Network management
    d) Hardware configuration
36. The data warehouse and operational environments are:
    a) The same
    b) Separated
    c) Interchangeable
    d) Both transactional

37. According to the twelve rules for a data warehouse, data warehouse data is:
    a) Mainly read-only with periodic batch updates
    b) Updated online in real time
    c) Deleted after each use
    d) Stored in flat files only
38. According to the twelve rules for a data warehouse, the data warehouse contains data with several levels of detail, including:
    a) Current detail data
    b) Old detail data
    c) Lightly and highly summarized data
    d) All of the above
39. The metadata in a data warehouse:
    a) Identifies and defines all data elements
    b) Provides the source, transformation, integration, storage, usage, relationships, and history of each data element
    c) Is a critical component of the data warehouse environment
    d) All of the above
40. In a star schema, facts and dimensions are normally represented by:
    a) Physical tables in the data warehouse database
    b) Flat files
    c) Network directories
    d) Log entries
41. In a star schema, fact and dimension tables are related by:
    a) Foreign keys
    b) Network addresses
    c) File names
    d) User IDs
42. Each dimension record in a star schema is related to:
    a) One fact record
    b) Thousands of fact records
    c) No fact records
    d) Only summary records
43. Which of the following is a type of BI tool?
    a) Dashboards and business activity monitoring
    b) Portals
    c) Data-mining tools
    d) All of the above
44. Dashboards use web-based technologies to:
    a) Present key business performance indicators in a single integrated view
    b) Store operational data

c) Manage network security
d) Process transactions

45. Portals provide:
    a) A unified, single point of entry for information distribution
    b) Transaction processing
    c) Data storage
    d) Network routing
46. Data-mining tools provide:
    a) Advanced statistical analysis to uncover problems and opportunities hidden within business data
    b) Basic reporting only
    c) Network management
    d) Hardware configuration
47. OLAP tools provide:
    a) Multidimensional data analysis
    b) Transaction processing
    c) Network security
    d) File storage
48. Data visualization tools provide:
    a) Advanced visual analysis and techniques to enhance understanding of business data
    b) Transaction processing
    c) Network routing
    d) Hardware configuration
49. In the evolution of BI, the 1970s were characterized by:
    a) Centralized reporting
    b) Spreadsheets
    c) Enterprise reporting
    d) Dashboards
50. In the evolution of BI, the 2010s to present are characterized by:
    a) Centralized reporting
    b) Spreadsheets
    c) Big Data analytics, Hadoop, NoSQL, and data visualization
    d) Enterprise reporting

Section B: True/False

1. Business intelligence (BI) describes a comprehensive, cohesive, and integrated set of tools and processes used to capture, collect, integrate, store, and analyze data.
2. The BI architecture is composed of people, processes, data, and technology.
3. ETL tools collect, filter, integrate, and aggregate internal and external data to be saved into a data store.
4. The data store in a BI framework is generally represented by a data warehouse or a data mart.
5. Query and reporting is the BI component that presents data to the end user in a variety of meaningful and innovative ways.
6. Data visualization helps the end user select the most appropriate presentation format, such as summary reports, maps, pie or bar graphs, and dashboards.
7. Data monitoring and alerting allow real-time monitoring of business activities.
8. Data analytics performs data analysis and data-mining tasks using the data in the data store.
9. The main goal of BI is improved decision-making.
10. BI provides a common data repository that fosters a single version of company data.
11. The precursor of the modern BI environment was the first-generation decision support system (DSS).
12. A DSS typically has a much broader focus and reach than a BI solution.
13. Operational data is mostly stored in multidimensional databases.
14. Decision support data gives tactical and strategic business meaning to operational data.
15. Operational data covers a short time frame.
16. Decision support data focuses on representing individual transactions.
17. Operational data has high granularity and a broad time span.
18. Decision support data is mostly used for updates.
19. Operational data typically has high-update volumes.
20. Decision support data typically has high query activity.
21. A decision support database is a specialized DBMS tailored to provide fast answers to complex queries.
22. The database schema for decision support must support complex data representations and contain aggregated and summarized data.
23. A data warehouse is an integrated, subject-oriented, time-variant, non-volatile collection of data that provides support for decision making.
24. Data warehouse data is arranged and optimized to provide answers to questions from diverse functional areas within a company.
25. Data warehouse data represents the flow of data through time.
26. Once data enters the data warehouse, it is never removed.

27. A data warehouse is usually a read-only database optimized for data analysis and query processing.
28. The ETL process stands for Extract, Transform, and Load.
29. A data mart is a large, enterprise-wide data warehouse.
30. A data mart provides decision support to a small group of people.
31. A star schema is a data modelling technique used to map multidimensional decision support data into a relational database.
32. A star schema has four components: facts, dimensions, attributes, and attribute hierarchies.
33. Facts are numeric measurements that represent specific business aspects or activities.
34. Dimensions are qualifying characteristics that provide additional perspectives to a given fact.
35. Attributes are used to search, filter, or classify facts.
36. Attribute hierarchies provide top-down data organization and the capability to perform drill-down and roll-up searches.
37. In a star schema, facts and dimensions are normally represented by physical tables in the data warehouse database.
38. Fact and dimension tables in a star schema are related by foreign keys.
39. Each dimension record in a star schema is related to thousands of fact records.
40. OLAP systems share three main characteristics: multidimensional data analysis techniques, advanced database support, and an easy-to-use end-user interface.
41. OLAP is an advanced data analysis environment that supports decision-making, business modelling, and operations research.
42. According to the twelve rules for a data warehouse, the data warehouse and operational environments are separated.
43. According to the twelve rules for a data warehouse, data warehouse data is mainly read-only with periodic batch updates from operational data.
44. According to the twelve rules for a data warehouse, the data warehouse development life cycle is process-driven.
45. The metadata in a data warehouse identifies and defines all data elements.
46. Dashboards use web-based technologies to present key business performance indicators in a single integrated view.
47. Portals provide a unified, single point of entry for information distribution.
48. Data-mining tools provide advanced statistical analysis to uncover problems and opportunities hidden within business data.
49. OLAP tools provide multidimensional data analysis.
50. Data visualization tools provide advanced visual analysis and techniques to enhance understanding of business data.

Section C: Matching Columns

Column A	Column B
1. Business intelligence (BI)	a) Collect, filter, integrate, and aggregate internal and external data to be saved into a data store
2. ETL tools	b) Comprehensive, cohesive, and integrated set of tools and processes used to capture, collect, integrate, store, and analyse data
3. Data store	c) Performs data selection and retrieval; used by data analysts to create queries and reports
4. Query and reporting	d) Optimized for decision support; generally represented by a data warehouse or data mart
5. Data visualization	e) Performs data analysis and data-mining tasks using the data in the data store
6. Data monitoring and alerting	f) Presents data to the end user in a variety of meaningful and innovative ways
7. Data analytics	g) Allows real-time monitoring of business activities; presents concise information in a single integrated view
8. Decision support system (DSS)	h) Arrangement of computerized tools used to assist managerial decision making
9. Operational data	i) Integrated, subject-oriented, time-variant, nonvolatile collection of data that provides support for decision making
10. Decision support data	j) Mostly stored in relational databases; optimized to support daily transactions
11. Data warehouse	k) Gives tactical and strategic business meaning to operational data
12. Data mart	l) Small, single-subject data warehouse subset
13. Star schema	m) Data modeling technique used to map multidimensional decision support data into a relational database
14. Facts	n) Numeric measurements that represent specific business aspects or activities
15. Dimensions	o) Qualifying characteristics that provide additional perspectives to a given fact
16. Attributes	p) Used to search, filter, or classify facts
17. Attribute hierarchies	q) Provide top-down data organization and capability to perform drill-down and roll-up searches

18. OLAP	r) BI style whose systems share multidimensional data analysis, advanced database support, and easy-to-use end-user interface
19. Dashboards	s) Use web-based technologies to present key business performance indicators in a single integrated view
20. Portals	t) Provide a unified, single point of entry for information distribution
21. Data-mining tools	u) Provide advanced statistical analysis to uncover problems and opportunities hidden within business data
22. OLAP tools	v) Provide multidimensional data analysis
23. Data visualization tools	w) Provide advanced visual analysis and techniques to enhance understanding of business data
24. Timespan	x) Area in which operational data differs from decision support data; covers a short time frame
25. Granularity	y) Area in which operational data differs from decision support data; must be presented at different levels of aggregation
26. Dimensionality	z) Area in which operational data differs from decision support data; focuses on representing individual transactions
27. Integrated	aa) Data warehouse characteristic; centralized, consolidated database that integrates data from multiple sources
28. Subject-oriented	ab) Data warehouse characteristic; data arranged and optimized to provide answers from diverse functional areas
29. Time-variant	ac) Data warehouse characteristic; data represents the flow of data through time
30. Non-volatile	ad) Data warehouse characteristic; once data enters, it is never removed

Section D: Short Questions

1. Describe the role of business intelligence in providing comprehensive business decision support. (4)
2. Describe the architecture, reporting styles, evolution, and benefits of business intelligence. (8)
3. Differentiate between operational data and decision support data. (6)
4. Identify the purpose, characteristics, and components of a data warehouse. (6)
5. Develop star and snowflake schemas for decision-making purposes. (6)
6. Describe the characteristics and capabilities of online analytical processing (OLAP). (6)
7. Describe the role and functions of data analytics and data mining. (4)
8. Explain how SQL analytic functions are used to support data analytics. (4)
9. Define data visualization and explain how it supports business intelligence. (4)
10. Explain the BI framework and its basic architectural components. (6)
11. Discuss the evolution of business intelligence from centralized reporting to modern BI environments. (6)
12. Explain the three main areas in which decision support data differs from operational data. (6)
13. Describe the three main requirements of a decision support database. (6)
14. Explain the four main characteristics of a data warehouse: integrated, subject-oriented, time-variant, and non-volatile. (8)
15. Describe the ETL process and its importance in data warehousing. (4)
16. Differentiate between a data warehouse and a data mart. (4)
17. Explain the components of a star schema: facts, dimensions, attributes, and attribute hierarchies. (8)
18. Discuss the twelve rules for a data warehouse. (6)
19. Explain the characteristics and capabilities of OLAP systems. (6)
20. Describe the various BI tools and their purposes. (8)